

Exact Lattice-Point Counting in a Disk in $\tilde{O}(R^{1/3})$ Time by a Stern–Brocot Convex-Hull Walk

Joseph Anointed
seth.cse@gmail.com

September 28, 2026

Abstract

We study the exact evaluation of the Gauss circle count $N(R) = \#\{(x, y) \in \mathbb{Z}^2 : x^2 + y^2 \leq R\}$ for integers R up to $4 \cdot 10^{18}$. The textbook method costs $\Theta(\sqrt{R})$ operations. We reduce the problem to a sum $\sum_{x=1}^m \lfloor \sqrt{R - x^2} \rfloor$ over a region where the boundary has slope in $[-1, 0]$, and evaluate it by walking along the upper convex hull of the lattice points under the arc, discovering hull edges by descending the Stern–Brocot tree and summing each edge in closed form. Only integer arithmetic is used. The implementation agrees with two independent oracles on every $R \leq 2 \times 10^6$, on 24,000 structured inputs up to $\approx 1.8 \times 10^{13}$, on 300 random inputs up to 9×10^{14} , and on seven large inputs up to 4×10^{18} . Measured work grows as $R^{0.338}$; at $R = 10^{18}$ it takes 27 ms versus 3.7 s for the $O(\sqrt{R})$ sweep. We prove correctness of the reduction, the edge summation, and the pruning rules; we do *not* prove the per-edge descent cost, which is supported here by measurement only. The technique belongs to a known family and we make no claim of novelty (Section 8).

1 Problem and baseline

Let $r = \text{isqrt}(R)$. Row-by-row summation gives

$$N(R) = 1 + 4r + 4 \sum_{x=1}^r \left\lfloor \sqrt{R - x^2} \right\rfloor,$$

which needs $r \approx \sqrt{R}$ integer square roots (or a two-pointer sweep). For $R = 10^{18}$ this is 10^9 steps. The bottleneck is that each column is visited individually. To do better one must count many columns at once.

2 Reduction to a low-slope arc

Let $m = \max\{m : 2m^2 \leq R\} = \text{isqrt}(\lfloor R/2 \rfloor)$, $f(x) = \lfloor \sqrt{R - x^2} \rfloor$, and $F = \sum_{x=1}^m f(x)$.

Lemma 1 (Octant reduction). *Let $T = \#\{(x, y) : x, y \geq 1, x^2 + y^2 \leq R\}$. Then $T = 2F - m^2$ and $N(R) = 1 + 4r + 4T$.*

Proof. The four axis half-lines contribute $4r$ points and the origin 1; the remaining points split into four congruent open quadrants of size T . For $1 \leq x \leq m$ we have $R - x^2 \geq R - m^2 \geq m^2$, so $f(x) \geq m$. The set $\{x \leq m\}$ contributes F points of T ; by the symmetry $x \leftrightarrow y$ so does $\{y \leq m\}$. Their intersection is $\{1 \leq x, y \leq m\}$, which lies in the disk because $2m^2 \leq R$, so it has m^2 points. If $x, y > m$ then $x^2 + y^2 \geq 2(m+1)^2 > R$, so nothing lies outside the union. Inclusion–exclusion gives $T = 2F - m^2$. $\square$

On $[0, m]$ the arc $g(x) = \sqrt{R - x^2}$ is concave and decreasing with $-1 \leq g'(x) \leq 0$, since $|g'(x)| = x/g(x) \leq 1$ when $2x^2 \leq R$. Consequently $f(c) - f(c+1) \in \{0, 1\}$ for consecutive columns.

3 Hull structure and closed-form edge sums

Let $K = \{(x, y) : x \leq m, x^2 + y^2 \leq R\}$ (convex) and let H be the convex hull of $K \cap \mathbb{Z}^2 \cap \{x \geq 0\}$.

Lemma 2 (Columns are read off the hull). *For every integer $c \in [0, m]$, the highest lattice point of K in column c is $(c, \lfloor h(c) \rfloor)$, where h is the upper boundary of H .*

Proof. $H \subseteq K$ since K is convex, so any lattice point of column c with $y \leq h(c)$ lies in K ; hence $f(c) \geq \lfloor h(c) \rfloor$. Conversely $(c, f(c)) \in H$ so $f(c) \leq h(c)$, and $f(c)$ is an integer. $\square$

Thus F equals the sum of $\lfloor h(c) \rfloor$ over columns, and h is piecewise linear with rational slopes. The slopes lie in $[-1, 0]$ by the remark above, so every edge direction is a primitive vector (dx, dy) with $0 \leq dy \leq dx$ meaning “right dx , down dy ”.

Lemma 3 (Edge sum). *For coprime $dx \geq 1, dy \geq 0$ and integer y ,*

$$\sum_{j=1}^{dx} \left\lfloor y - \frac{j dy}{dx} \right\rfloor = dx y - C, \quad C = \frac{dx dy + dx + dy - 1}{2}.$$

Hence k consecutive steps of the same vector starting at (x, y) contribute $k dx y - dx dy \binom{k}{2} - kC$ to F .

Proof. $\lfloor y - a \rfloor = y - \lceil a \rceil$. For coprime (dx, dy) , $\sum_{j=1}^{dx-1} \lfloor j dy/dx \rfloor = (dx-1)(dy-1)/2$, and $\lfloor j dy/dx \rfloor = \lfloor j dy/dx \rfloor + 1$ for $1 \leq j < dx$, while the term $j = dx$ equals dy . So $\sum_j \lfloor j dy/dx \rfloor = \frac{(dx-1)(dy-1)}{2} + (dx-1) + dy = C$. The k -step formula follows by summing over starting points $(x + i dx, y - i dy)$. $\square$

4 Finding the edges: Stern–Brocot descent

Write $\sigma = dy/dx \in [0, 1]$ for the steepness of a direction. From a hull vertex P , the next edge has the *smallest* steepness among directions v with $P + v \in K$. Directions with steepness in $[0, 1]$ are the nodes of the Stern–Brocot tree between $0/1 = (1, 0)$ and $1/1 = (1, 1)$.

Lemma 4 (Reach is an interval and is monotone). *For $P \in K$ and steepness $\sigma \in [0, 1]$, $\{t \geq 0 : P + t(1, -\sigma) \in K\} = [0, T_\sigma]$, and T_σ is non-decreasing in σ .*

Proof. Interval: K is convex. Monotone: lowering σ moves the ray up; the lower boundary $-g$ is never reached because along the ray $y \geq x - 1 \geq -1$ for lattice-adjacent points (Section 2 slope bound), so leaving K can only happen through the upper arc, and a ray that is higher leaves no later. $\square$

Lemma 5 (Pruning). *Let L, R_t be Stern–Brocot neighbours ($\det = 1$), with $P + L \notin K$, every direction flatter than L having no point in K , and $P + R_t \in K$. Let $M = L + R_t$. (a) If $P + M \notin K$, then every direction flatter than or equal to M has no point in K ; replace $L \leftarrow M$. (b) If $P + M \in K$, replace $R_t \leftarrow M$ (pushing the old one on the stack). (c) Let $k = \max\{k : P + k R_t \in K\}$. Any direction strictly between L and R_t that has a point in K has $dx \leq (k+1) dx(R_t) - 1$. So if $dx(L) + dx(R_t)$ exceeds this bound, R_t is the flattest usable direction.*

Algorithm 1: HullWalk(R): computes $F = \sum_{x=1}^m \lfloor \sqrt{R - x^2} \rfloor$

Input: integer $R \geq 1$
Output: F and m

```

1  $m \leftarrow \text{isqrt}(\lfloor R/2 \rfloor)$ ;  $(x, y) \leftarrow (0, \text{isqrt}(R))$ ;  $F \leftarrow 0$ ; stack  $S \leftarrow [(1, 1), (1, 0)]$  (top =  $(1, 0)$ );
2 while  $x < m$  do
3    $(dx, dy) \leftarrow \text{top}(S)$ ;  $k \leftarrow \text{MaxSteps}(x, y, dx, dy)$ ; // exponential + binary search
4    $F \leftarrow F + k dx y - dx dy \binom{k}{2} - kC$ ;  $x \leftarrow x + k dx$ ;  $y \leftarrow y - k dy$ ;
5    $L \leftarrow \text{pop}(S)$ ; if  $x \geq m$  then stop;
6   while top( $S$ ) leaves  $K$  from  $(x, y)$  do
7      $L \leftarrow \text{pop}(S)$ 
8   repeat
9      $R_t \leftarrow \text{top}(S)$ ;  $M \leftarrow L + R_t$ ;
10    if  $x + dx(M) > m$  then break;
11    if  $(x, y) + M \in K$  then
12      push  $M$ 
13    else
14       $k' \leftarrow \text{MaxSteps}$  for  $R_t$  (cached);
15      if  $dx(M) + dx(R_t) > (k' + 1)dx(R_t) - 1$  then break;
16       $L \leftarrow M$ 
17  until stop;
18 return  $F$ 

```

Proof. (a) Directions in the subtree between L and M have $dx \geq dx(L) + dx(M) > dx(M)$, and by Lemma 4 their reach is at most $T_{\sigma(M)} < dx(M)$, so they have no lattice point in K . (b) is the definition of the search. (c) Any v strictly between L and R_t is flatter than R_t , so $dx(v) \leq T_{\sigma(v)} \leq T_{\sigma(R_t)} < (k + 1)dx(R_t)$. $\square$

Theorem 1 (Correctness). *Algorithm 1 returns F exactly, hence $N(R) = 1 + 4r + 4(2F - m^2)$.*

Proof sketch. Invariant: at the top of each iteration (x, y) is the top lattice point of column x (a hull vertex, or the start), the sum of f over columns $1..x$ has been added, and no direction flatter than the stack top's predecessor chain has a point in K . Lemma 5 shows each descent move preserves the invariant and that on termination the top of the stack is the flattest direction with a point in K , i.e. the next hull edge (it exists because $(1, 1)$ always stays inside while $x < m$: $f(c + 1) \geq f(c) - 1$). Lemma 3 adds exactly the column tops along the edge, which by Lemma 2 equal f . Adjacency ($\det = 1$) of consecutive stack entries is preserved by pushes of mediants. $\square$

Remark 1. A tempting shortcut is false: the predicate “ $L + jR_t$ is inside” is *not* monotone in j (it is out, then possibly in, then out again as dx outgrows the reach), so a plain binary search over j is invalid. The algorithm therefore steps one Stern–Brocot node at a time and relies on Lemma 5(c) to stop.

5 Complexity

Every hull edge costs one MaxSteps call, $O(\log R)$ predicate evaluations. The number of hull edges is at most the number of vertices of the convex hull of the integer points of a disk of radius $\sqrt{R}$,

which is $\Theta(R^{1/3})$ by the classical result of Balog and Bárány [1]. The octant walk uses about $1/8$ of the circle. What we do *not* prove is that the total number of Stern–Brocot descent steps is $O(R^{1/3} \log R)$; the standard amortization for hyperbola versions is folklore and we did not rederive it. Section 6 measures it.

6 Experimental validation

Code: `gauss.py` (reference, Python integers) and `gauss.cpp` (C++17, `_int128` for $x^2 + y^2$ and sums), single core, -O2. Oracles: (A) $O(\sqrt{R})$ two-pointer sweep; (B) $N = 1 + 4 \sum_{k \geq 0} (\lfloor R/(4k+1) \rfloor - \lfloor R/(4k+3) \rfloor)$, used in Python for $R \leq 3000$.

Correctness results (all zero mismatches).

- Python vs. both oracles, every $R \in [0, 3000]$; vs. oracle A, every $R \in [3001, 120000]$ and 300 random $R \leq 10^{10}$.
- C++ vs. oracle A, every $R \in [0, 2 \cdot 10^6]$.
- C++: 24,000 structured inputs a^2 , $a^2 \pm 1$, $a^2 + b^2$, $a^2 + b^2 - 1$, $2a^2$ ($a, b \leq 3 \cdot 10^6$), plus 300 random $R \in [10^{11}, 9 \cdot 10^{14}]$.
- $R = 10^k + 12345$ for $k = 6, 8, \dots, 18$ and $R = 4 \cdot 10^{18}$.

R	fast (s)	sweep (s)	predicate calls	hull edges	calls/ $R^{1/3}$
10^6	0.00001	0.000	572	44	5.7
10^8	0.00002	0.000	2,742	210	5.9
10^{10}	0.00007	0.000	13,502	921	6.3
10^{12}	0.00030	0.005	64,675	4,313	6.5
10^{14}	0.00182	0.038	301,730	20,117	6.5
10^{16}	0.00616	0.409	1,410,596	93,259	6.5
10^{18}	0.02748	3.691	6,553,646	432,570	6.6

Table 1: Measured cost for $R = 10^k + 12345$. Every value matched the sweep.

Performance. A log–log least-squares fit gives exponents 0.338 for predicate calls and 0.332 for hull edges, consistent with $R^{1/3}$. The ratio calls/ $R^{1/3}$ is nearly flat (5.7 to 6.6 over twelve decades), so a $\log R$ factor is not visible in this range. The edge count is about $0.43 R^{1/3}$. At $R = 4 \cdot 10^{18}$: $N = 12,566,370,614,358,945,969$ from both methods (this exceeds the signed 64-bit range, hence 128-bit accumulation). As a sanity check on the asymptotics, $N(10^{18} + 12345) - \pi R = -24,688$, far below $\sqrt{R} = 10^9$.

7 Limitations and threats to validity

- The $O(R^{1/3} \log R)$ bound for total descent work is empirical, over $R \leq 4 \cdot 10^{18}$, not proved.
- Testing is exhaustive only to $2 \cdot 10^6$; beyond that it is sampled. Structured inputs target boundary cases but cannot rule out a rare failure.
- The proofs of Lemma 4 and Theorem 1 are sketched at the level above; a machine-checked proof was not attempted.

- The C++ code assumes $R < 2^{63}$ for the input and uses 128-bit for products; larger R needs wider types.
- Timings are single runs on one machine, not averaged.

8 Prior art and what is new

Counting lattice points under a convex curve with a Stern–Brocot walk is known for the hyperbola (divisor summatory function) and appears in the competitive-programming and computational number theory literature. Its application to the circle is a direct adaptation. This paper’s contribution is an explicit derivation for the disk (octant reduction, column-top lemma, closed-form edge sums, a sound pruning rule), a correction of one plausible-looking but invalid shortcut (Remark 1), and a test record. I did not search the literature exhaustively; the algorithm should be assumed to be known.

9 Method notes

Two errors were caught during development, both by reasoning before testing: (i) the claim that “inside” is monotone along $L + jR_t$ (false, Remark 1); (ii) a stopping rule based only on the remaining x -range $m - x$, which is correct but makes each edge cost up to $O(\sqrt{R})$; Lemma 5(c) replaces it with a local reach bound. The first version of the code that implemented these rules passed all tests.

References

- [1] A. Balog and I. Bárány, On the convex hull of the integer points in a disc, in *Proc. 7th Ann. Symp. Computational Geometry*, ACM, 1991, 162–165.